17.04.25

OPENCAMPUS.sh

Machine Learning With TensorFlow

GENERAL INTRODUCTION

- Introduction of the Instructors
- Intro to opencampus.sh
- Organizational Matters
- Introductory Discussion on AI
- Coursera Registration
- Course Projects
- ML Frameworks

INTRODUCTION OF THE INSTRUCTORS

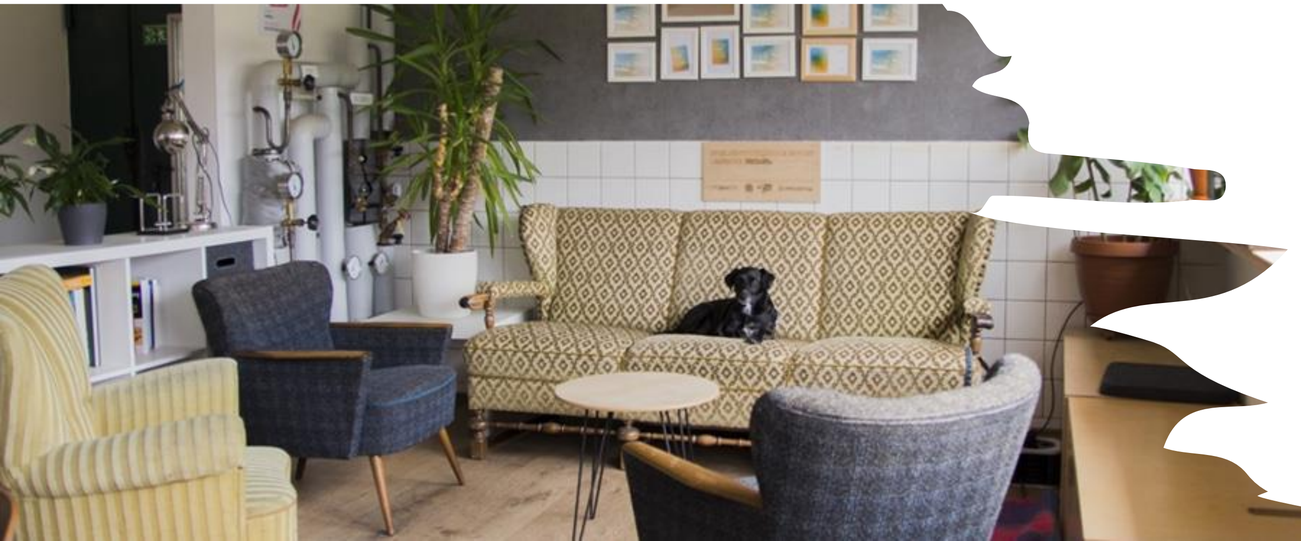
OPENCAMPUS.sh

- Nonprofit organization which oversees a variety of initiatives
- Offering a wide range of educational opportunities, support, and networking for entrepreneurs, creatives, and anyone curious, regardless of age, educational background, or origin
- The services are open to everyone and mostly free.
- The goal is to support the entrepreneurial landscape, promote creative change processes, and contribute to innovative and sustainable future development.









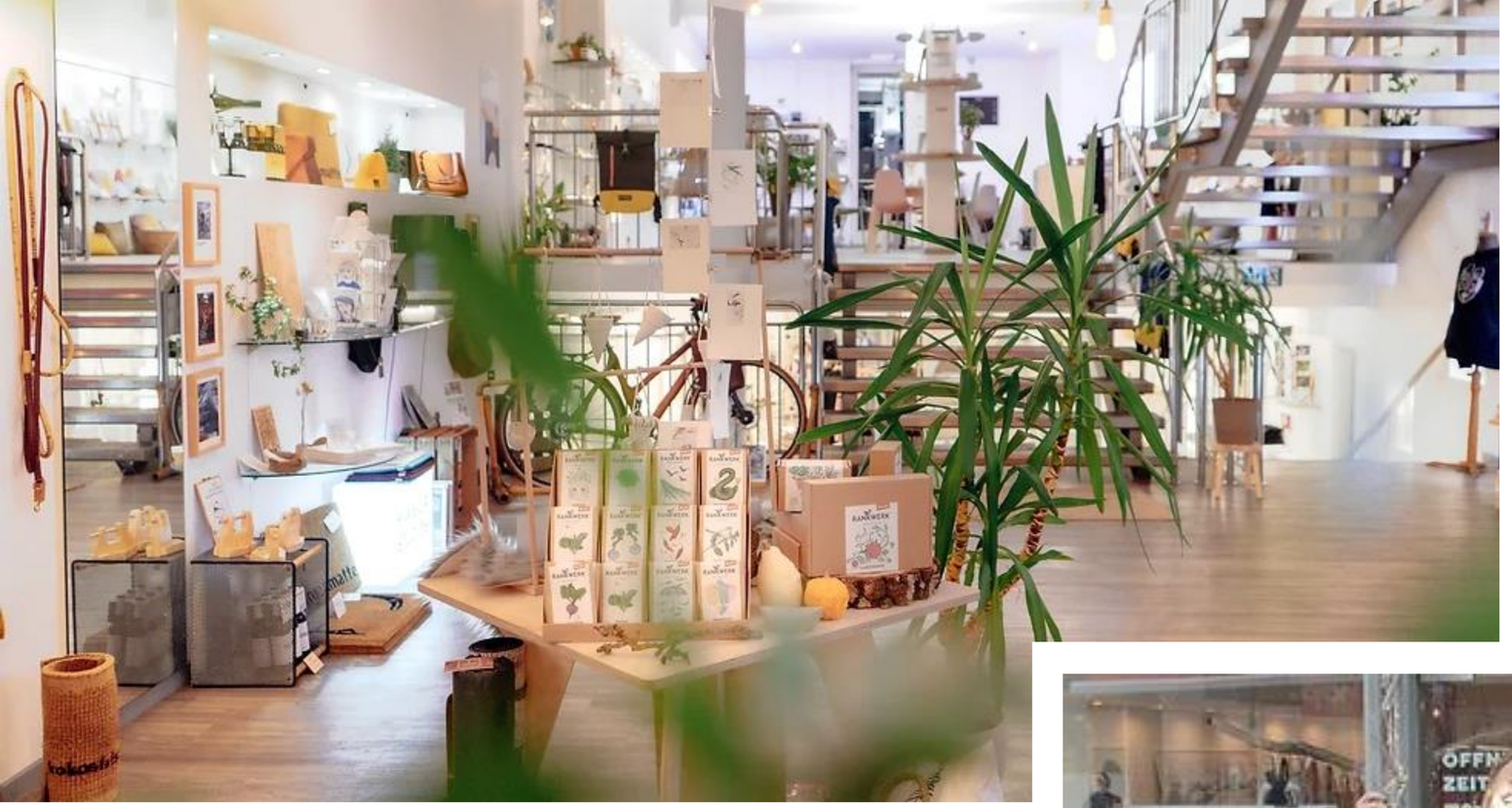
starter kitchen.de



CoBL

COZY WORKING, CULTURE & EVENTS





KOSMOS
by opencampus.sh





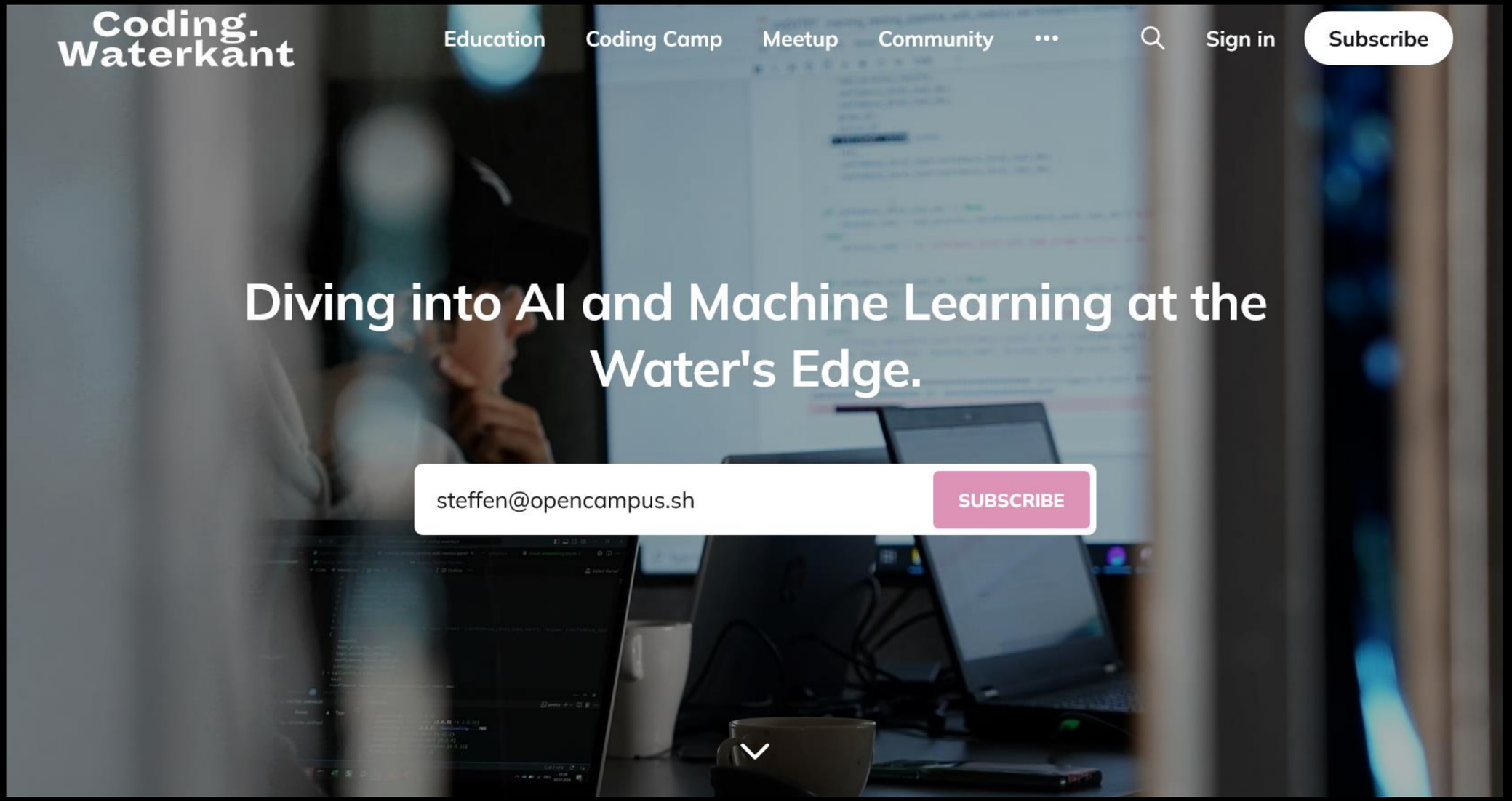






FABLAB KIEL


Visual Science
Communication
Research
WAS SIND OZEANE WERT?
HOW MUCH ARE OCEANS WORTH?
WATERKANT
FESTIVAL
June 19-20



https://coding.waterkant.sh



Coding. Waterkant 2025

Coding.Waterkant 2025

July 07 - 11

- Work for four consecutive days:
  - on your own machine learning project or
  - support others in their project
- Take part in workshops
- Get input and feedback by invited experts
- Present your work to a larger audience.
- Take educational leave
- Get accommodation on site

Meetup

Events suchen | Kiel, DE

Neue Gruppe starten - 30% Rabatt!

PRO Kostenlos ausprobieren

Vernetzungen

Im Rahmen der ständigen Verbesserungen unserer Plattform haben wir die E-Mail-Adresse deiner Gruppe in coding-waterkant-announce@messages.meetup.com geändert. Du kannst dies in den optionalen Einstellungen deiner Gruppe ansehen.



Coding.Waterkant

★★★★½ (126)

Neu: Event-Feedbackübersicht

Tippe auf die Sterne, um alle deine Eventbewertungen in der Übersicht anzuzeigen.

Weitere Informationen

Kiel, Deutschland

981 Mitglieder · Öffentliche Gruppe

Organisiert von **opencampus.sh** and **6 others**

Teilen:

Info | Events | Mitglieder | Fotos | Diskussionen | Mehr

Event erstellen | Gruppe verwalten

Über uns

Our meetup is organized with support of opencampus.sh and the Digitale Wirtschaft Schleswig-Holstein (DiWiSH).

Mehr lesen



Organizers

opencampus.sh and **6 others**

Nachricht

Members (981)

Alles ansehen



ORGANIZATIONAL MATTERS

Attendance Registration:

Online

- **Use your full names in the zoom meetings!**
- **Turn your camera on!**

Presence

- **Scan the QR-Code if you participate in presence**

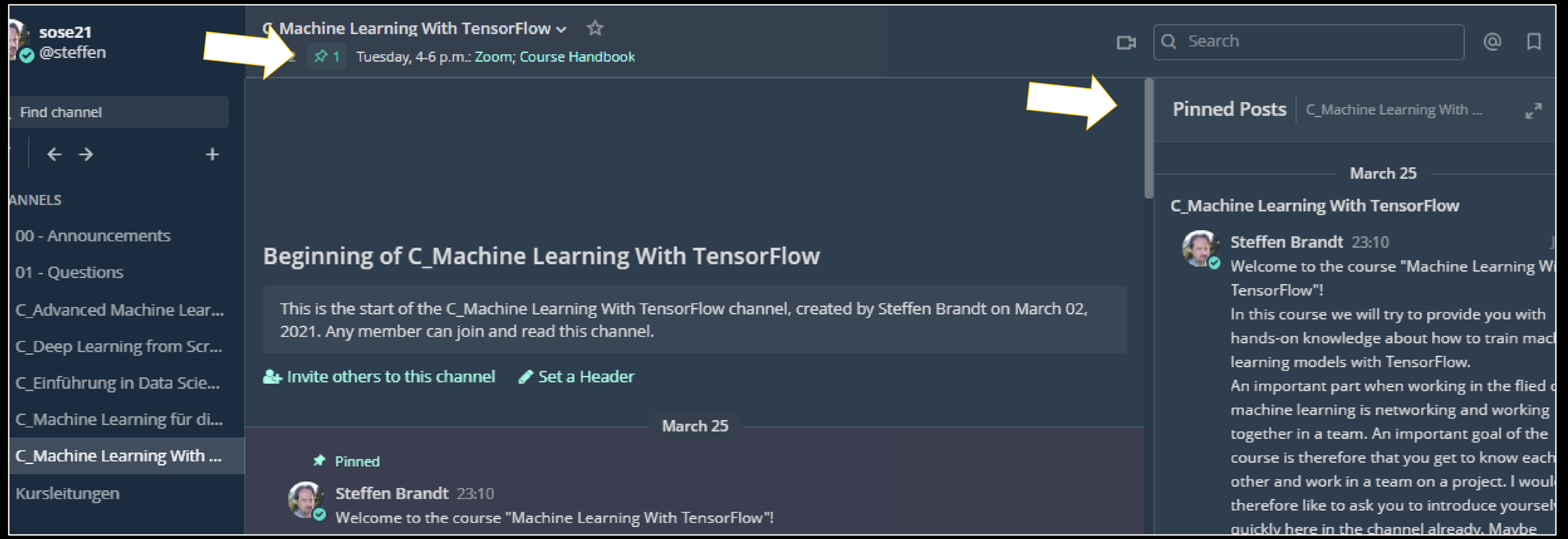
Please write us if you will not go on with the course!

MATTERMOST CHAT

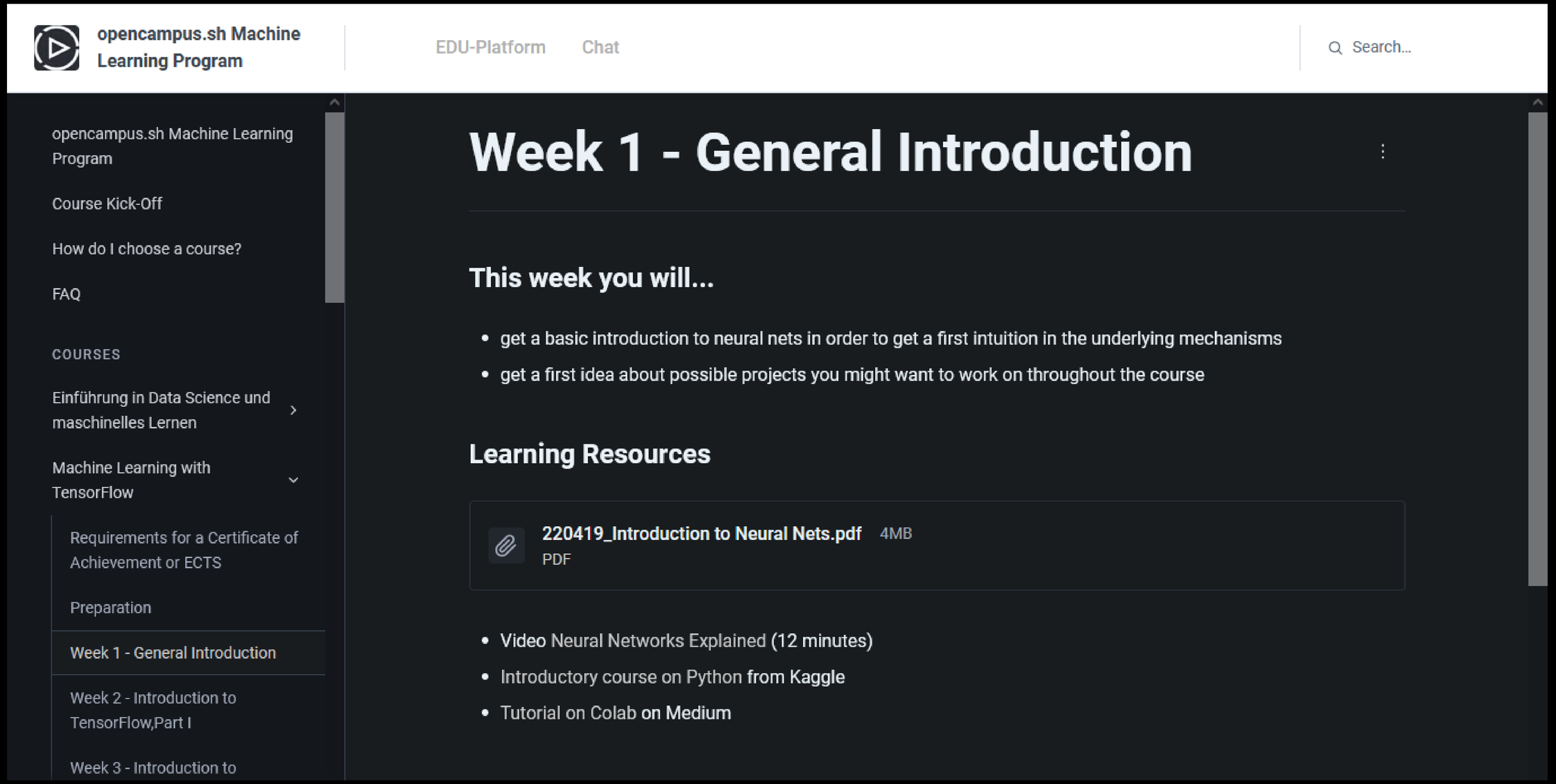
- Complete your profile in the Mattermost chat with your full name and a photo.

- Please, always ask any questions to us in the course chat!!

MATTERMOST CHAT



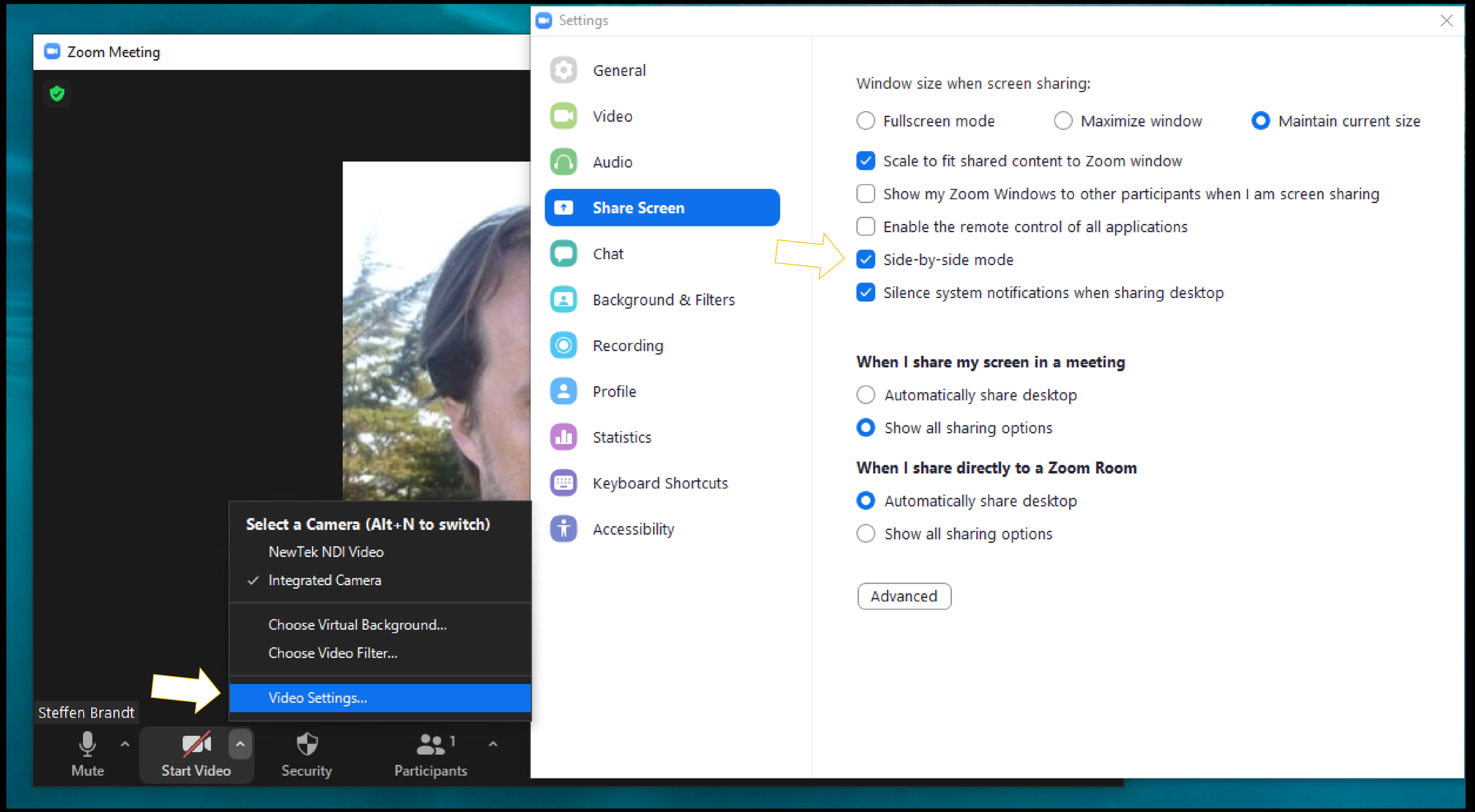
COURSE HANDBOOK



ZOOM

- Try the different viewing modes:
  - Gallery View/ Active Speaker
  - Split Screen/ Full Screen Mode

- Maybe watch this video to get an idea: https://www.youtube.com/watch?v=v3IPAbpVjd4


Zoom Meeting
Select a Camera (Alt+N to switch)
NewTek NDI Video
Integrated Camera
Choose Virtual Background...
Choose Video Filter...
Video Settings...
Steffen Brandt
Mute
Start Video
Security
Participants
1
Settings
General
Video
Audio
Share Screen
Chat
Background & Filters
Recording
Profile
Statistics
Keyboard Shortcuts
Accessibility
Window size when screen sharing:
Fullscreen mode
Maximize window
Maintain current size
Scale to fit shared content to Zoom window
Show my Zoom Windows to other participants when I am screen sharing
Enable the remote control of all applications
Side-by-side mode
Silence system notifications when sharing desktop
When I share my screen in a meeting
Automatically share desktop
Show all sharing options
When I share directly to a Zoom Room
Automatically share desktop
Show all sharing options
Advanced

17.04.2025
18:15 - 20:00
General Introduction
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



Jeremy von Winckelmann
speaker



Steffen Brandt
speaker
LinkedIn-Profil

24.04.2025
18:15 - 20:00
Introduction to TensorFlow for AI, Machine Learning, and Deep Learning, Part I
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



Jake Petersen
speaker

08.05.2025
18:15 - 20:00
Introduction to TensorFlow for AI, Machine Learning, and Deep Learning, Part II
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



Hannes Körner
speaker

15.05.2025
18:15 - 20:00
Convolutional Neural Networks, Part I
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



Jake Petersen
speaker

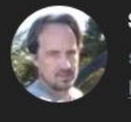
22.05.2025
18:15 - 20:00
Convolutional Neural Networks, Part II
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



Hannes Körner
speaker

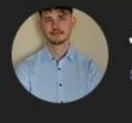
Coding. Waterkant

05.06.2025
18:15 - 20:00
Natural Language Processing, Part I
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



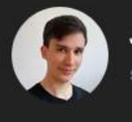
Steffen Brandt
speaker
LinkedIn-Profil

12.06.2025
18:15 - 20:00
Natural Language Processing, Part II
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



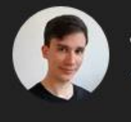
Jake Petersen
speaker

19.06.2025
18:15 - 20:00
Sequences, Time Series and Prediction, Part I
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



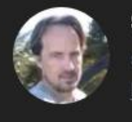
Jeremy von Winckelmann
speaker

26.06.2025
18:15 - 20:00
Sequences, Time Series and Prediction, Part II
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



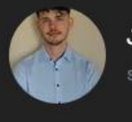
Jeremy von Winckelmann
speaker

03.07.2025
18:15 - 20:00
Presentation of the Final Projects, Part I
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE

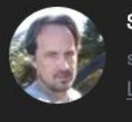


Steffen Brandt
speaker
LinkedIn-Profil

10.07.2025
18:15 - 20:00
Presentation of the Final Projects, Part II
Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE



Jake Petersen
speaker



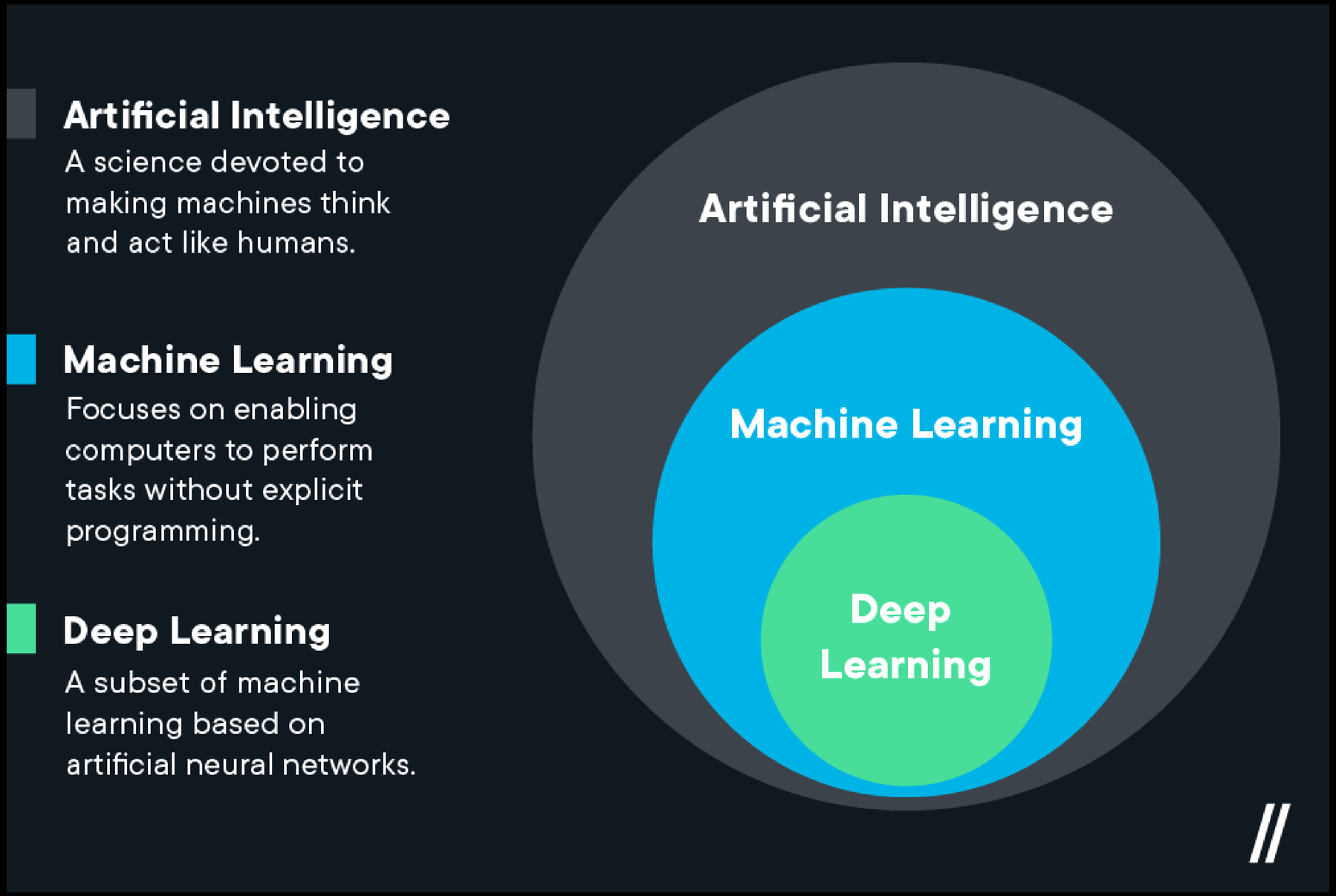
Steffen Brandt
speaker
LinkedIn-Profil

PROJECTS MILESTONES

24.04.	Project Pitches
01.05.	Further Project Proposals and Discussions in Mattermost
08.05.	Form Groups
15.05.	Literature Review (*Submission Deadline*: 18.05.)
22.05.	Dataset Characteristics
05.06.	Definition of Model Evaluation
12.06.	Baseline Model Estimation (*Submission Deadline*: 15.06.)
22.06.	Individual Feedback Sessions
03.07.	Project Presentations, Part I
10.07.	Project Presentations, Part II

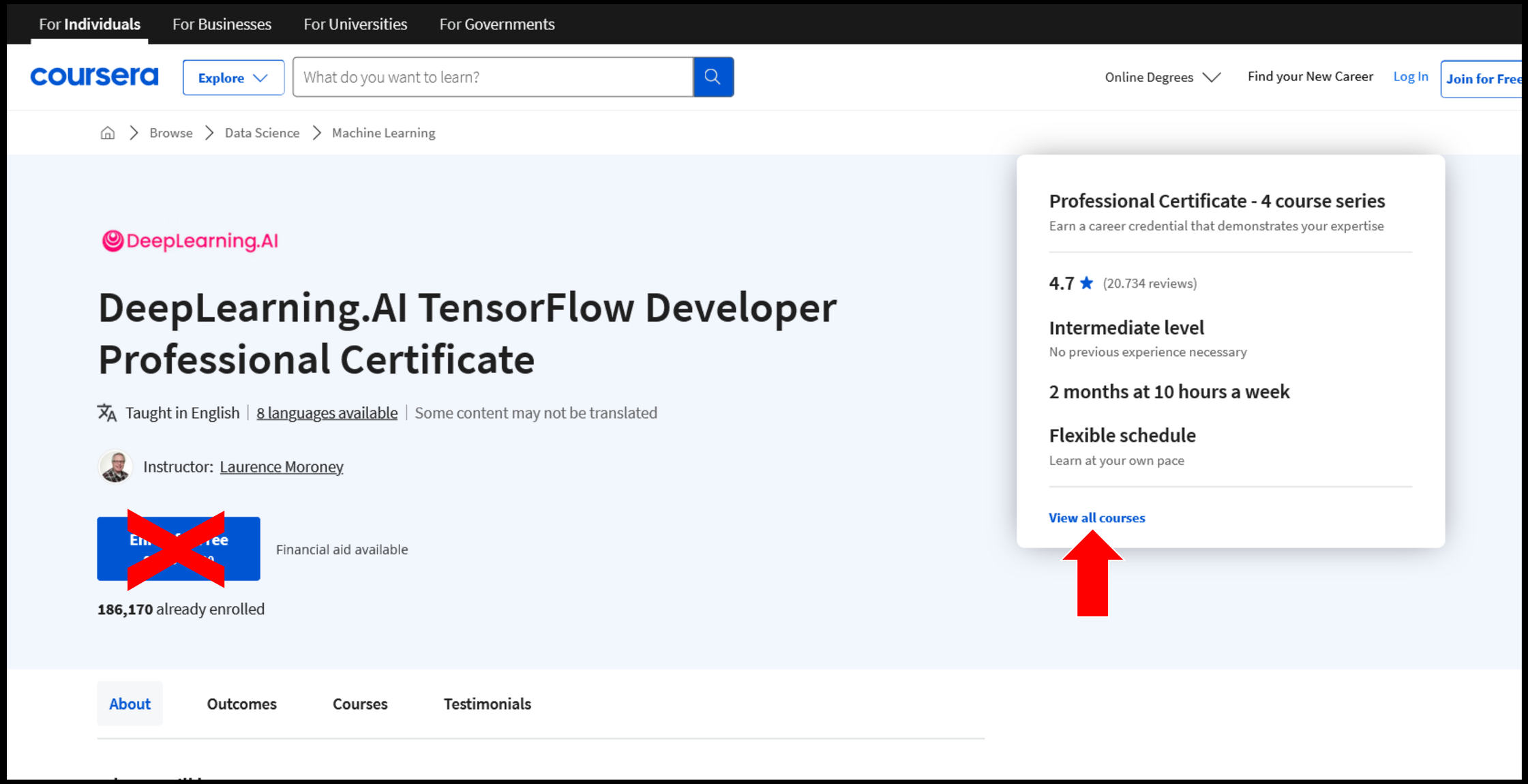
FIRST BREAKOUT

- 15-20 Minutes
- Present yourself
- Discussion Questions:
  - Do you know examples for Machine Learning?
  - Do you know examples for Deep Learning?



Deep Learning vs. Machine Learning—What's the Difference? (2021, February 8). Flatiron School. https://flatironschool.com/blog/deep-learning-vs-machine-learning/

DO NOT SIGN UP FOR THE SPECIALIZATION



CLICK ON THE INDIVIDUAL COURSE

Professional Certificate - 4 course series

TensorFlow is one of the most in-demand and popular open-source deep learning frameworks available today. The DeepLearning.AI TensorFlow Developer Professional Certificate program teaches you applied machine learning skills with TensorFlow so you can build and train powerful models.

Read more



Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning

Course details

Course 1 • 17 hours • **4.8** ★ (18.883 ratings)

What you'll learn

- ✓ Learn best practices for using TensorFlow, a popular open-source machine learning framework
- ✓ Build a basic neural network in TensorFlow
- ✓ Train a neural network for a computer vision application
- ✓ Understand how to use convolutions to improve your neural network

Skills you'll gain

Computer Vision | Tensorflow | Machine Learning

Convolutional Neural Networks in TensorFlow

Course 2 • 16 hours • **4.7** ★ (7.912 ratings)

Instructor

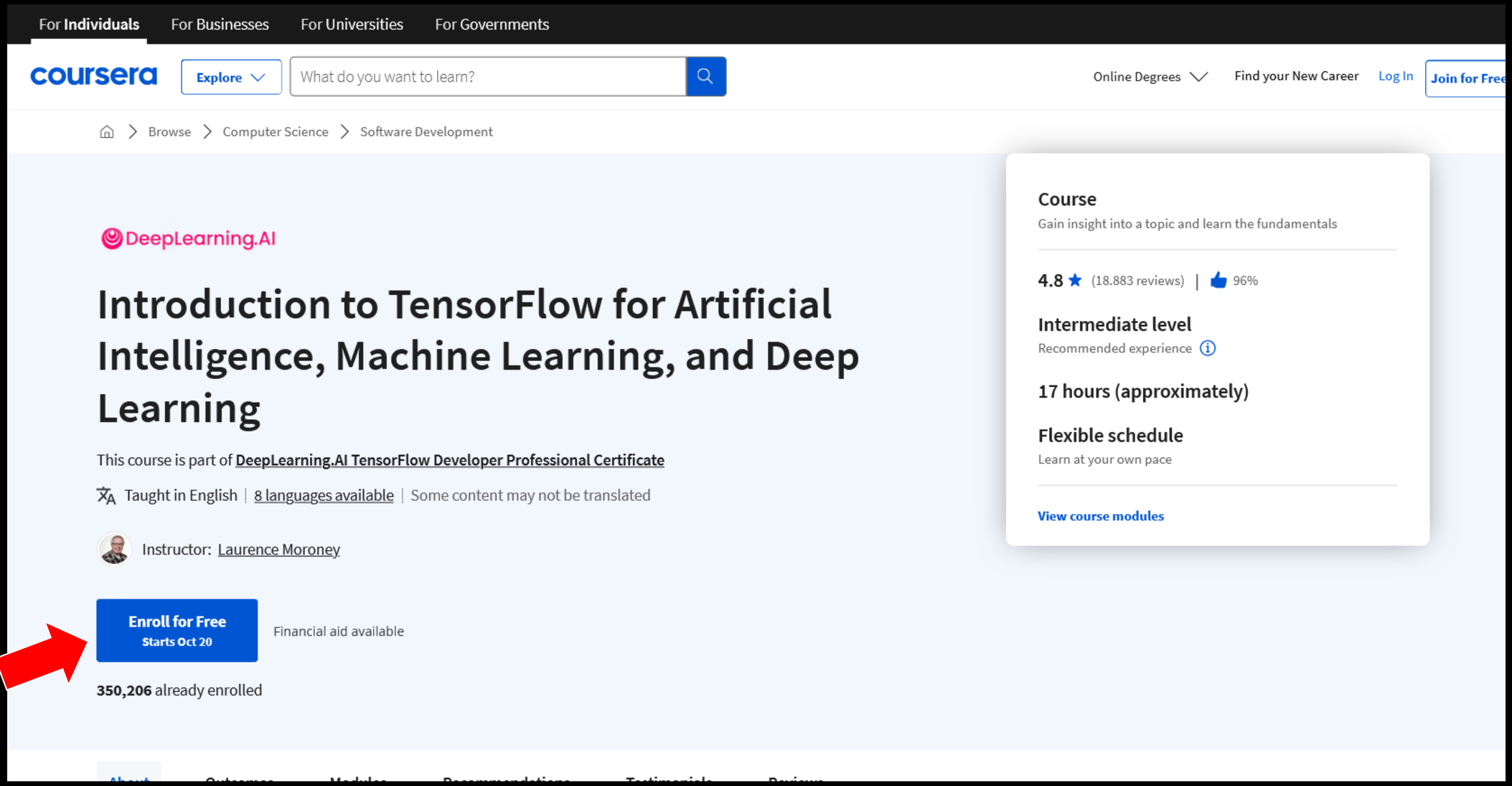


Laurence Moroney
DeepLearning.AI
15 Courses • 456,481 learners

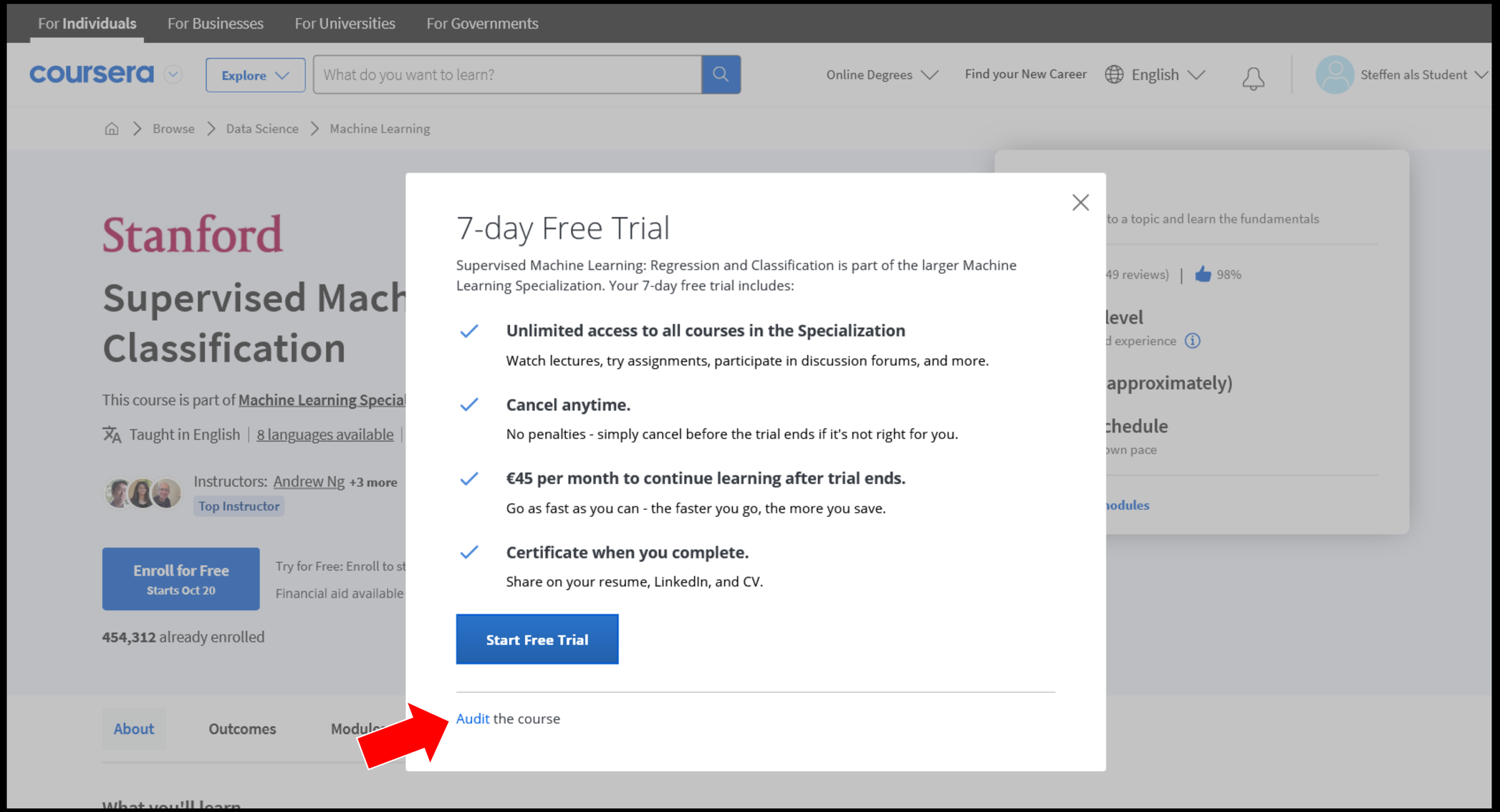
Offered by

DeepLearning.AI
Learn more

ENROLL FOR INDIVIDUAL COURSE



SELECT AUDIT MODUS



EXERCISES

- Each week two to four of you will present the exercises given in the course handbook
- Each of you presents at least once

EXERCISES:
WHO WILL PRESENT NEXT WEEK?

PROJECTS

- Bring your own idea and data
- First look for a problem, then look for data
- Good approach is to look for time consuming or boring tasks humans are dealing with

RELEVANCE OF THE PROJECTS

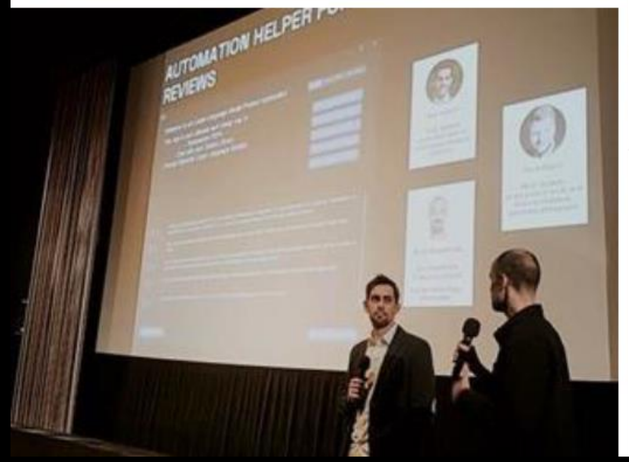
- Most important for a career in ML will be work experience and your GitHub profile
- Focus on building a noteworthy GitHub project repository
- Use the template repository
- Outstanding projects will be nominated for the VDE Machine Learning Prize, which is awarded once a year

VDE

SPECIAL PRIZE MACHINE LEARNING













COURSES

Fine-Tuning and Deployment of Large Language Models

Archive

EVENTS

Coding.Waterkant 2023

Prototyping Week

PROJECTS

How to Start, Complete, and Submit Your Project

Possible Projects

Past Projects

ADDITIONAL RESOURSES

Glossary

Coursera

Selecting the Optimizer

Choosing the Learning Rate

Learning Linear Algebra

Learning Python

Support Vector Machines

ML Statistics

TOOLS

Git

RStudio

Google Colab

Zoom

Powered by GitBook

How to Start, Complete, and Submit Your Project

Starting Your Project

Working on Your Project

Submitting Your Project

Was this helpful?

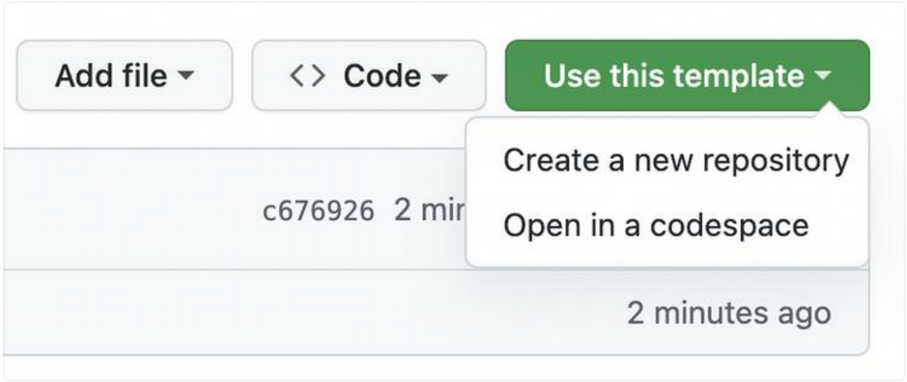
Export as PDF

In all Machine Learning courses you have:

- to complete a machine learning project in a team of up to 4 participants,
- attend at least all but 2 sessions of the course, and
- use the provided project template repository for documentation (unless otherwise instructed).

Starting Your Project

1. **Navigate to the** Template Repository
2. **Use this Template**: Above the file list, click the "Use this template" button.



Use this template button

3. **Create Repository from Template**: You'll be prompted to name your new repository and you can choose whether it should be public or private. You'll also have the option to include all branches in the template repository, if there are more than one.
4. **Create Repository**: Click "Create repository from template" to create the new repository.
5. **Clone the New Repository**: You can now clone the new repository to your local machine using `git clone` and start working on your project.

Working on Your Project

PROJECT INTERESTS

ChallengeData By MathA

HOME CHALLENGES FAQ CONTACT LOGIN EN



AGIR POUR L'ÉDUCATION
UN ENJEU SCIENTIFIQUE POUR LA SOCIÉTÉ

Welcome to the Challenge Data website!!

Each year, we organize machine learning challenges from data provided by public services, companies and laboratories: **general documentation** and **FAQ**. Seasons begin in January; the challenges are introduced in the context of Stéphane Mallat's lesson at the Collège de France.

A prize ceremony for the best participants of the preceding season will be held in February at the College de France (03/02/2022).

For participants

Guide to create an account, choose your challenges and submit solutions.

ML LIBRARIES (LOWER LEVEL)







PYTORCH

ML LIBRARIES (HIGHER LEVEL)



Transformers

spaCy



haystack
by deepset



LangChain

LUDWIG



Auto gpt

DEVELOPMENT ENVIRONMENTS











TASKS UNTIL NEXT WEEK

- Completion of the learning material of week 1 and 2 of the course "introduction to TensorFlow“
- Complete the two assignments given in the following notebooks:
  - Assignment Notebook 1
  - Assignment Notebook 2
- Bring questions!